EMPLOYEE TRAINING & DEVELOPMENT ANALYTICS

PT PLN UNIT INDUK WILAYAH NUSA TENGGARA TIMUR

CREATED BY

Hosea Yurie Haganta S Meliala

Data Analyst | Former HR Intern – PT PLN UIW NTT

A data-driven case study of Training Participation, Learning Outcomes, Employee Development, Training Completion & Training Investment.

Power BI Portfolio Project | Synthetic Dataset | 2026

00 — Contents

01 — Executive Overview

Employee training data contains more information than simple participation records. When properly structured and analyzed, it can provide visibility into participation, training exposure, learning outcomes, completion performance, and training investment.

This project develops an end-to-end HR Analytics solution for Employee Training & Development using Microsoft Power BI. The solution transforms employee and training records into an integrated analytical framework that supports Monitoring → Analysis → Diagnostic Investigation → Management Decision Support.

> The project is inspired by the author's previous experience during an HR internship at PT PLN Unit Induk Wilayah Nusa Tenggara Timur (UIW NTT). The dataset used is synthetic and created for portfolio and analytical demonstration purposes.

Analytical Area	Primary Question
Training Participation	How widely do training programs reach employees?
Learning Outcomes	Do assessment scores improve from pre-training to post-training?
Employee Development	Which groups show relatively lower cumulative training exposure?
Completion	Which areas show relatively lower completion performance?
Training Investment	Where is training spending concentrated?
Diagnostic Analysis	Which areas deserve further investigation?

02 — Project Context

Training and development requires organizational resources, employee participation, and financial investment. Reporting only the number of employees trained does not provide a complete picture of training performance.

> **This project therefore evaluates training from multiple dimensions:**
> **Participation → Exposure → Completion → Learning → Investment → Diagnostic Analysis.**

A 97% participation rate can indicate broad coverage, but it does not necessarily mean every employee receives the same amount of training. Likewise, positive post-training assessment does not automatically demonstrate long-term employee performance or business impact.

03 — Business Problem

- How widely do training programs reach the employee population?
- Which departments and employee groups have relatively lower training exposure?
- Do assessment scores improve from pre-training to post-training?
- Which training categories or departments show relatively lower completion performance?
- How much is invested in training and where is investment concentrated?
- Which areas require further investigation based on participation, exposure, completion, learning, and investment?

A centralized Power BI solution was developed to transform training records into an interactive and consistent analytical view for evidence-based HR discussion and management review.

04 — Business Objectives

Objective	Definition
Participation	Measure employee training participation and organizational coverage.
Effectiveness	Evaluate measured learning improvement using pre- and post-training assessment scores.
Development	Identify differences in cumulative training exposure across employees, departments, and job levels.
Completion	Identify areas with relatively lower training completion performance.
Investment	Analyze total training cost, cost per participant, spending distribution, and trends.
Decision Support	Translate analytical findings into management-oriented recommendations and investigation areas.

05 — Analytical Framework

01 — Business Understanding

02 — Data Preparation

03 — Data Modeling

04 — KPI & DAX Development

05 — Visual Analytics

06 — Business Interpretation

07 — Diagnostic Analysis

08 — Recommendations

The workflow is designed to move from raw data toward a decision-support layer rather than stopping at dashboard production.

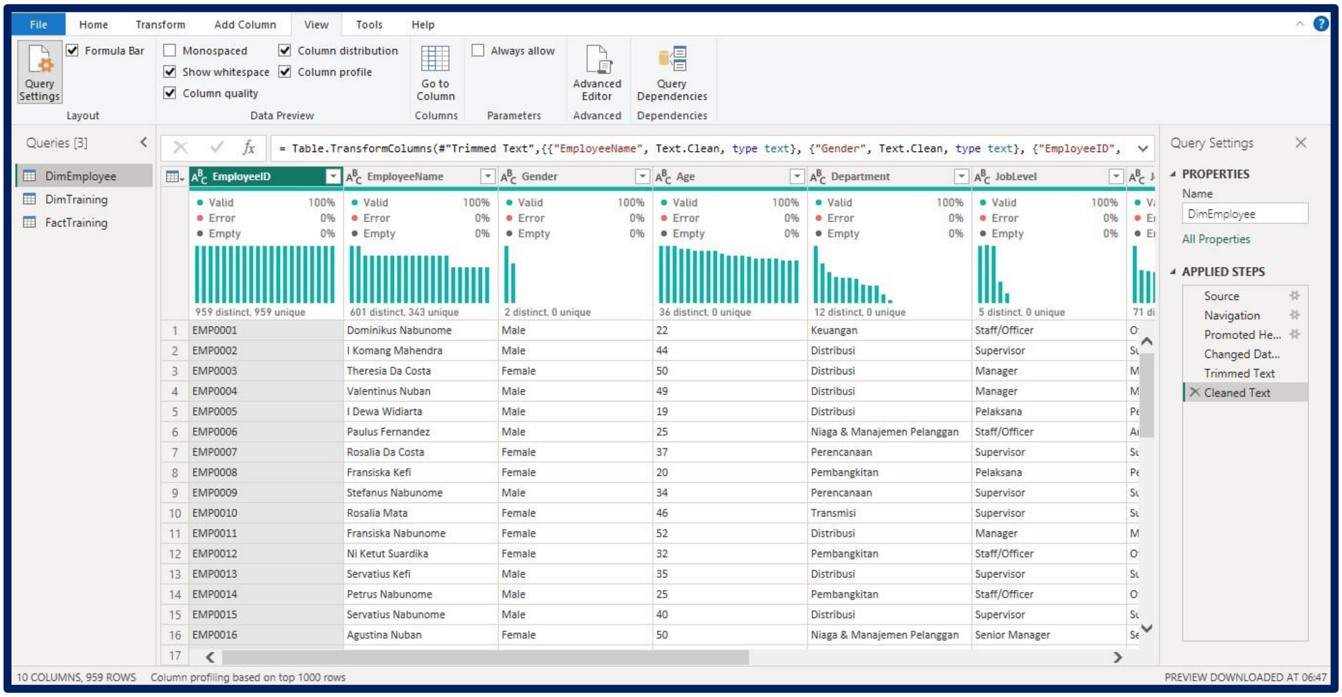
06 — Dataset & Data Structure

Data Component	Volume / Role
Employees	959 employee records
Training Records	Approximately 4,000 training participation records
Training Programs	25 training programs
Core Sheets	Employee, Training Transaction, Training Master
Date Dimension	Dedicated time dimension for year and trend analysis

Employee data provides organizational context; training transactions provide participation, hours, assessment scores, completion status, and cost; the training master provides program attributes; and the Date dimension supports time analysis.

07 — Data Preparation | Power Query

- Standardize data types for dates, numeric values, training hours, scores, cost, and identifiers.
- Review missing values, invalid values, category consistency, and identifier integrity.
- Standardize Department, Job Level, Training Category, Training Type, and Completion Status.
- Prepare employee and training keys for relationships.
- Create and validate a dedicated Date dimension for time analysis.



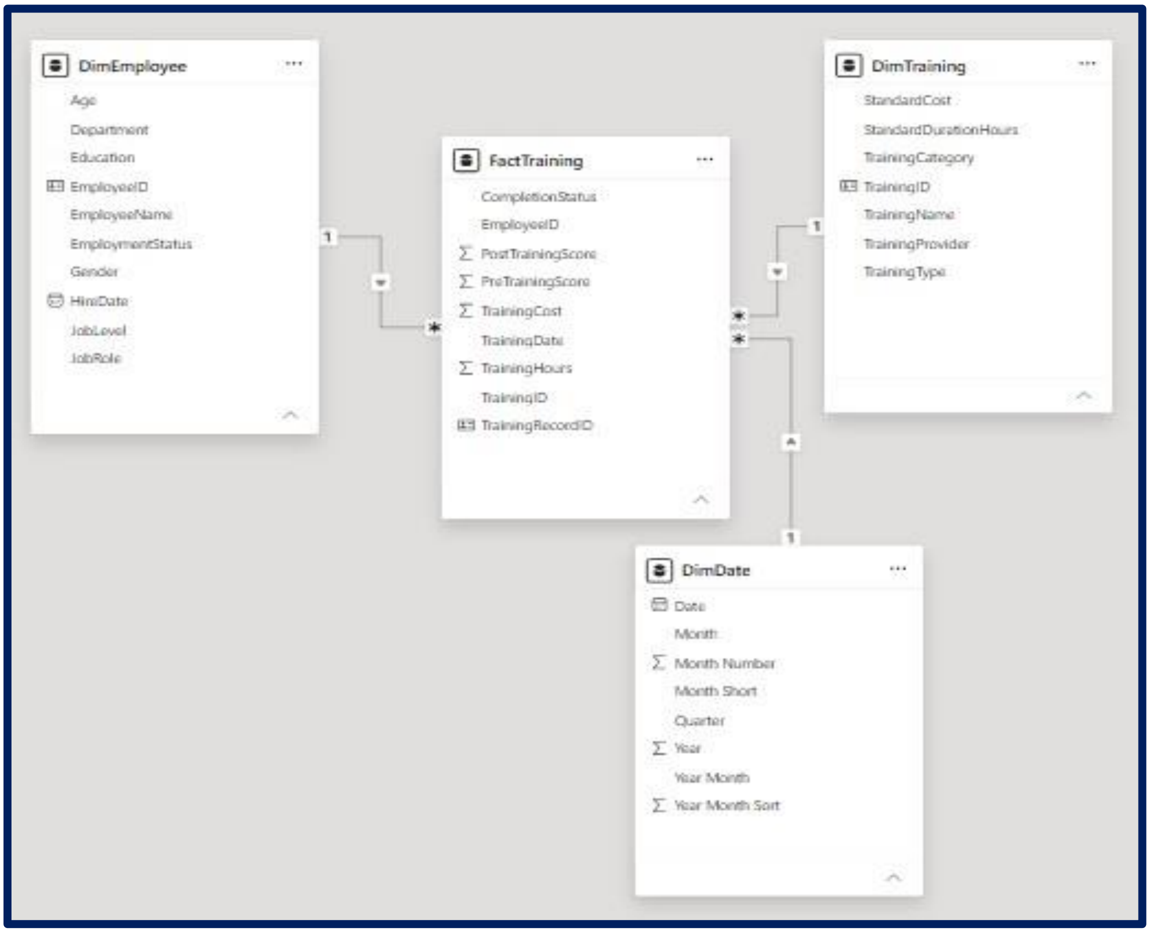
(Picture: Power Query Step)

08 — Data Modeling | Star Schema

The project uses a Star Schema to separate descriptive dimensions from measurable training transactions.

> **Dimensions describe the training context, while the fact table stores measurable training activity.**

Model Component	Analytical Role
FactTraining	Stores measurable training activity and outcomes.
DimEmployee	Provides employee and organizational context.
DimTraining	Provides training program and category context.
DimDate	Provides consistent time filtering and trend analysis.

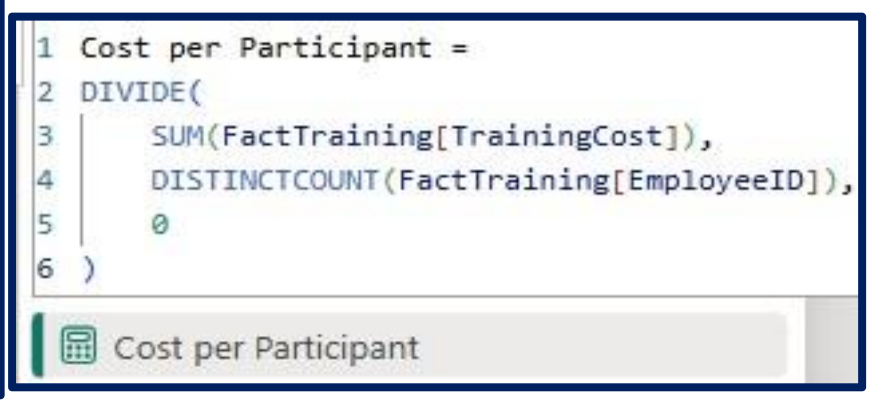


(Picture: Data Modelling Step)

09 — DAX Measures & KPI Layer

Domain	Measure	Purpose
Workforce	Total Employees	Count unique employees
Participation	Training Participants	Count unique participating employees
Participation	Training Participant Rate	Participants ÷ employees
Development	Total Training Hours	Sum training hours
Development	Average Training Hours	Average training exposure
Development	Employees Below 30 Hours	Identify lower exposure
Effectiveness	Average Pre-Training Score	Baseline score
Effectiveness	Average Post-Training Score	Post-training score
Effectiveness	Learning Improvement	Post – Pre
Completion	Training Completion Rate	Completed records ÷ total records
Investment	Total Training Cost	Sum training cost
Investment	Cost per Participant	Total cost ÷ participants

(Picture: DAX Step)



(Picture: Example DAX Formula Step)

10 — KPI Definitions & Analytical Governance

KPI	Definition
Total Employees	Distinct employee count.
Training Participants	Distinct employees with at least one training participation record.
Training Participant Rate	Training Participants ÷ Total Employees.
Total Training Hours	Total recorded training hours.
Average Training Hours	Average cumulative training exposure among analyzed participants.
Employees Below 30 Hours	Employees whose cumulative training hours are below the analytical threshold.
Learning Improvement	Average Post-Training Score − Average Pre-Training Score.
Training Completion Rate	Completed training records ÷ total training records.
Total Training Cost	Sum of recorded training costs.
Cost per Participant	Total Training Cost ÷ Training Participants.

> **The 30-hour threshold is an analytical segmentation rule created for this portfolio project. It is not an official PLN policy, mandatory training requirement, or validated minimum development standard.**

11 — Power BI Dashboard Architecture

Page	Primary Purpose
01 — Training Overview	Participation and workforce coverage
02 — Training Effectiveness	Learning outcomes and completion
03 — Employee Development	Training exposure and development distribution
04 — Training Investment	Training cost and investment distribution
05 — Executive Summary	Management-level consolidation
06 — Training Insights	Diagnostic and drill-down analysis

> **Analytical Flow:**
> **Operational Visibility → Performance Analysis → Diagnostic Analysis → Executive Decision Support**

12 — Training Overview

Element	Configuration
Filters	Year • Department • Training Category
KPIs	Total Employees • Training Participants • Training Participant Rate • Total Training Hours • Total Training Cost
Visuals	Training Participants by Department
Analytical Question	How broadly does training reach the employee population, and how is participation distributed across departments?

(Picture: Training Overview Dashboard)

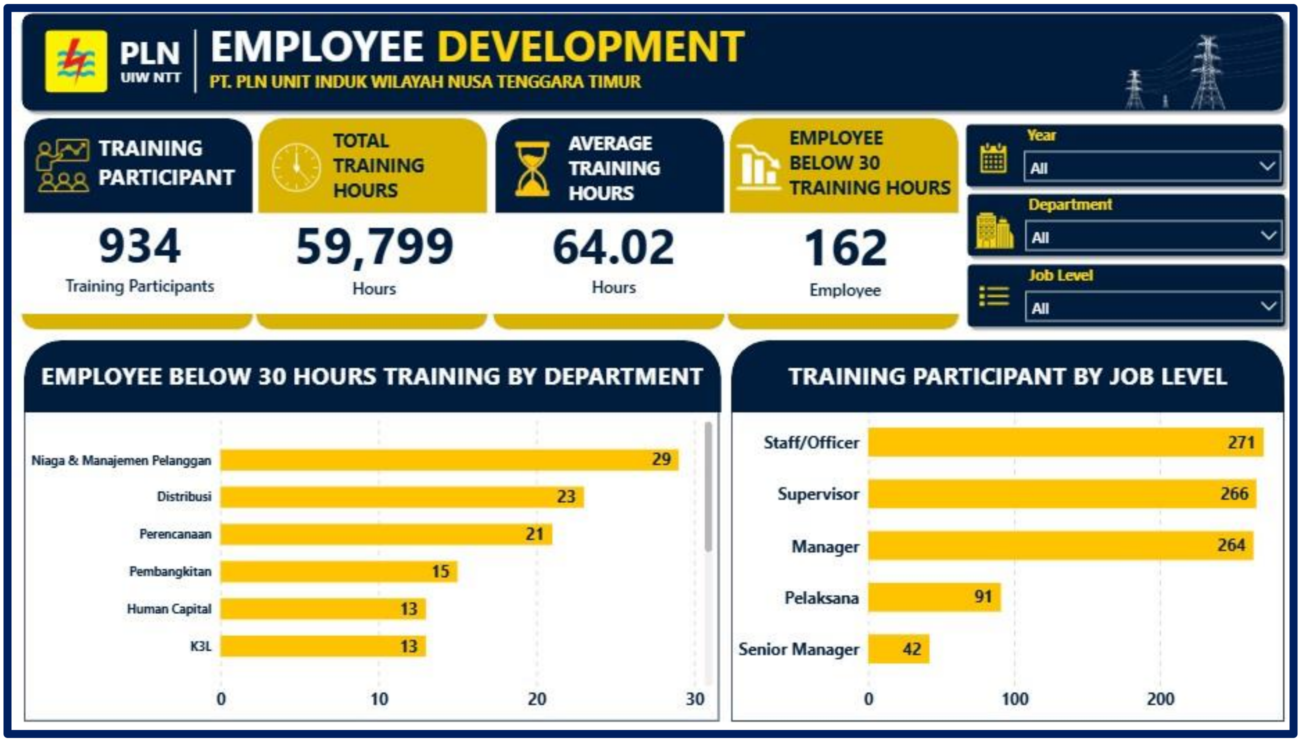
13 — Training Effectiveness

Element	Configuration
Filters	Year • Department • Training Category
KPIs	Average Pre-Training Score • Average Post-Training Score • Learning Improvement • Training Completion Rate
Visuals	Average Pre vs Post Score by Training Category • Completion Rate by Training Category
Analytical Question	Do assessment scores improve after training, and which categories require further review of completion performance?

(Picture: Training Effectiveness Dashboard)

14 — Employee Development

Element	Configuration
Filters	Year • Department • Job Level
KPIs	Training Participants • Total Training Hours • Average Training Hours • Employees Below 30 Training Hours
Visuals	Employees Below 30 Hours by Department • Training Participants by Job Level
Analytical Question	Does broad participation translate into relatively balanced training exposure across employees and organizational groups?



(Picture: Employee Development Dashboard)

15 — Training Investment

Element	Configuration
Filters	Year • Department • Training Category
KPIs	Total Training Cost • Cost per Participant
Visuals	Training Cost by Training Category • Cost per Participant by Department • Training Cost Trend • Training Cost by Department
Analytical Question	Where is training investment concentrated, and how does expenditure vary across departments and categories?

(Picture: Training Investment Dashboard)

16 — Executive Summary

Element	Configuration
Filters	Executive-level filters as configured
KPIs	Participation • Learning • Development • Completion • Investment
Visuals	Management KPI and key finding views
Analytical Question	Which findings are most important for management attention?

(Picture: Executive Summary Dashboard)

17 — Business Insights

17.1 – High Training Coverage

Approximately 97.4% of employees participated in training. This indicates broad training reach, but participation does not necessarily mean equal training exposure.

17.2 – Positive Measured Learning Improvement

Average learning performance increased by approximately +7.99 points from pre-training to post-training assessment. This is a measured learning indicator, not proof of long-term business impact.

17.3 – Training Exposure Gap

162 employees are below 30 cumulative training hours, approximately 16.9% of the modeled workforce. This is a diagnostic segmentation, not proof of insufficient training.

17.4 – Training Investment Concentration

K3 / OHS represents approximately 52.3% of total modeled training investment. High concentration is not automatically inefficient; it should be interpreted against program requirements and outcomes.

17.5 – Strongest Observed Category-Level Learning Improvement

Governance / Risk records approximately +8.23 points. This is a candidate for deeper program review, not proof that the category is causally superior.

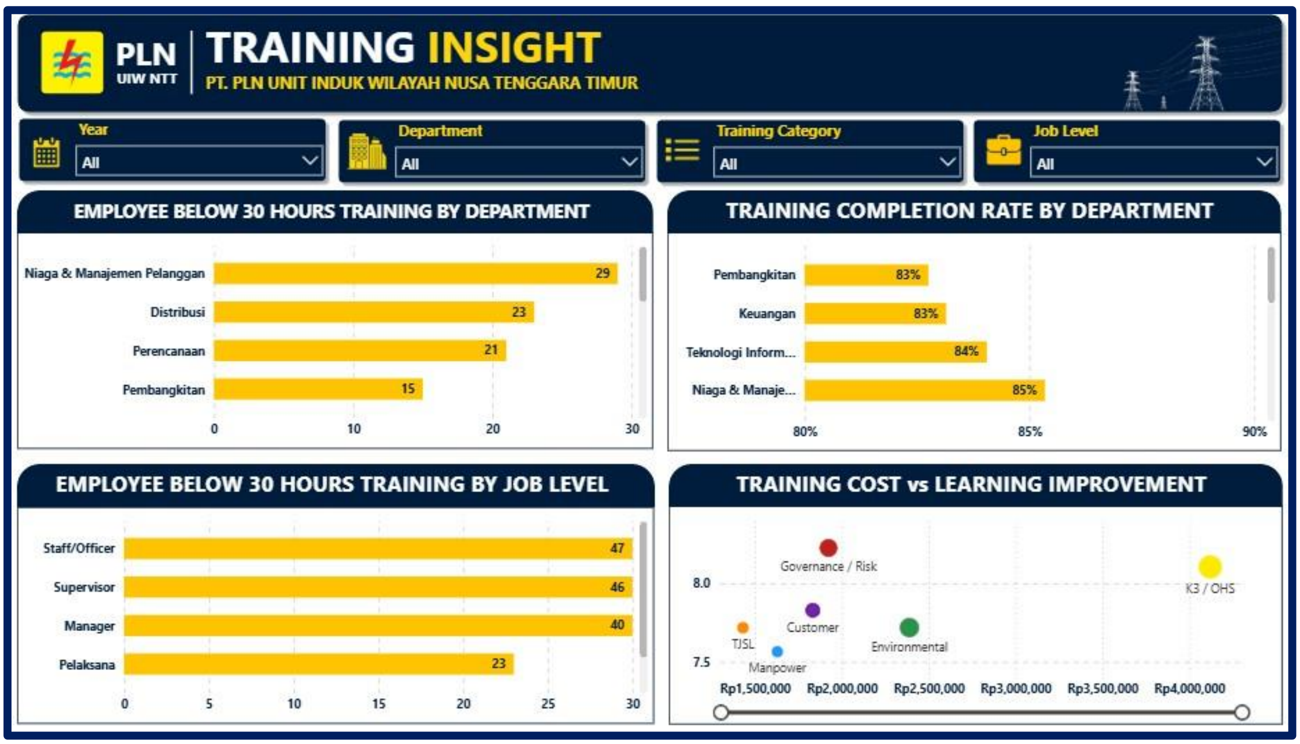
17.6 – Completion Performance Gap

TJSL records an observed completion rate of approximately 81.6%. It should be treated as a diagnostic signal requiring further investigation.

18 — Training Insights | Diagnostic Analysis

The four core dashboards answer “What is happening?”. The Training Insights page extends the analysis toward “Where are potential gaps concentrated?” and “What should be investigated further?”

Diagnostic View	Purpose
Employees Below 30 Hours by Department	Identify departments with larger counts of lower training exposure.
Employees Below 30 Hours by Job Level	Identify exposure differences across organizational levels.
Training Completion Rate by Department	Compare completion performance and locate potential diagnostic signals.
Training Cost vs Learning Improvement	Compare investment intensity and measured learning improvement by training category.



(Picture: Training Insight Dashboard)

> Training Cost vs Learning Improvement is not a Training ROI calculation. It provides a comparative basis for identifying programs that may deserve deeper review.

19 — Business Recommendations

01 — Close Training Exposure Gaps

Monitor employees below the analytical threshold and drill down by Department, Job Level, Job Role, Tenure, and Training Category. Use the result to identify development groups requiring review.

02 — Improve Training Completion

Review lower completion areas such as TJSL by Department, Training Program, Training Type, Duration, and Participant Group before introducing corrective actions.

03 — Monitor High-Investment Programs

Evaluate Total Training Cost, Cost per Participant, Participant Volume, Completion Rate, and Learning Improvement together to improve investment transparency.

04 — Learn From Stronger Learning Outcomes

Review program characteristics behind stronger measured learning outcomes, including duration, delivery method, assessment design, participant profile, completion, provider, and objectives.

05 — Establish a Training Performance Framework

Monitor Participation + Exposure + Completion + Learning + Investment rather than relying on one KPI.

20 — Management Action Priority

Priority	Management Focus	Next Analytical Step
1	Training Exposure	Investigate groups with consistently lower exposure.
2	Completion	Investigate categories/departments with relatively lower completion.
3	Investment	Review high-investment programs against participant and learning metrics.
4	Learning Outcomes	Identify practices from categories with stronger measured learning outcomes.

21 — Analytical Limitations

01 — Synthetic Dataset

The dataset was created for portfolio and analytical demonstration purposes and should not be interpreted as actual PT PLN UIW NTT performance.

02 — Learning Improvement ≠ Business Impact

Pre/post assessment improvement does not directly measure productivity, operational efficiency, revenue, cost savings, job performance, behavior change, or long-term impact.

03 — No Causal Inference

The analysis identifies descriptive and comparative patterns and does not establish that training caused observed improvement.

04 — No Financial Training ROI

The dataset does not contain sufficient business-performance variables to calculate financial training ROI.

05 — 30-Hour Threshold

The 30-hour threshold is an analytical segmentation rule created for this project, not an official minimum training requirement.

06 — Category-Level Interpretation

Rates and averages should be interpreted together with participant volume, record count, duration, and program characteristics.

22 — Future Analytical Development

The current project focuses on Training Activity + Learning Outcomes + Investment. Future development could incorporate additional HR and business data.

- Employee performance ratings
- Productivity indicators
- Promotion and career progression
- Training satisfaction and feedback
- Training objectives
- Actual budget
- Operational KPIs
- Post-training performance

Current: Training Activity Analytics → Next: Training Outcome Analytics → Future: Training Impact Analytics → Advanced: Training ROI Analysis

23 — Project Deliverables

Area	Deliverable
Power BI	Training Overview
Power BI	Training Effectiveness
Power BI	Employee Development
Power BI	Training Investment
Power BI	Executive Summary
Power BI	Training Insights
Data & Analytics	Power Query transformations
Data & Analytics	Star-schema model and Date dimension
Data & Analytics	Reusable DAX measures and KPI definitions
Documentation	HR Analytics Case Study

24 — Final Conclusion

The Employee Training & Development Analytics project demonstrates how employee and training records can be transformed into an integrated HR Analytics solution using Microsoft Power BI.

> The analytical journey is:
> Participation → Exposure → Completion → Learning → Investment → Diagnostic Analysis.

The modeled dataset shows 934 training participants, approximately 97.4% participation, approximately +7.99 points measured learning improvement, 162 employees below 30 cumulative training hours, approximately 52.3% of training investment concentrated in K3 / OHS, and approximately 81.6% completion rate for TJSL.

The central HR Analytics lesson is that high participation alone does not guarantee balanced employee development. A more complete training evaluation requires consideration of who participates, how much exposure they receive, whether training is completed, whether measured learning improves, how investment is distributed, and which areas require further investigation.

The project therefore moves beyond basic training reporting toward a structured HR decision-support framework.

25 — Author's Note

This project was developed as a portfolio case study inspired by the author's previous experience as an HR intern at PT PLN Unit Induk Wilayah Nusa Tenggara Timur (UIW NTT).

The project applies an end-to-end analytics workflow covering business problem definition, data preparation, data modeling, DAX development, Power BI visualization, diagnostic analysis, business interpretation, and recommendation development.

The dataset is synthetic and was created for portfolio purposes. The project demonstrates how Data Analytics and HR Analytics concepts can be applied within a realistic organizational context while maintaining a clear distinction between analytical demonstration and actual organizational performance.